

Lesson 6: Security Risks Part 2

45 minutes

Overview

The lesson begins with a review of security risks by watching a video on Cybersecurity & Crime. Following this, the class does an investigation into the Equifax breach, and what went wrong. The class ends with a Kahoot quiz to review security risks.

Standards

Full Course Alignment

CSP Conceptual Framework

- **IOC-2** - The use of computing innovations may involve risks to your personal safety and identity.

CSTA K-12 Computer Science Standards (2017)

- **NI** - Networks & the Internet

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Assessment: Check For Understanding

Objectives

Students will be able to:

- Confidently explain security risks and their impact on society
- Describe the role human error played in the Equifax breach

Preparation

- Listen to the podcast segments and practice starting and stopping at the marked places
- Get the Kahoot quiz set up and ready to go for the Wrap Up
- Check the "[Teacher's Lounge](#)" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our [Virtual Lesson](#)

Modifications

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Kahoot Wrap Up Quiz](#) - Resource
- [Security Risks Part 2](#) - Slides

▼ Make a Copy

For the students

- [CSP U8L6 - Optional Podcast Transcripts](#) - Activity Guide
 - [Equifax Data Breach: What Went Wrong \(Podcast\)](#) - Resource
- (Download)

- [Planet Money: Bad Credit Bureau \(Podcast\)](#) - Resource
- [The Internet: Cybersecurity and Crime](#) - Video ([Download](#))

Teaching Guide

Warm Up (5 minutes)

Remarks

Yesterday we investigated a few security risks. Let's watch a video together to review some of those security risks.

 Video: [The Internet: Cybersecurity & Crime](#)

Activity (35 minutes)

Remarks

When we think about security, we often think about mistakes that we personally have made - like clicking on a link in a text message from someone we don't know. But what about companies? Security is a major concern for companies, and sometimes human errors can have massive consequences.

Today we are going to take a look at the credit reporting bureau Equifax. In 2017, Equifax was hacked and the private information of around 145 million people was compromised. But what is Equifax and what data was stolen?

We are going to listen to two segments of two different podcasts. In the first, we will hear the history of credit bureaus leading up to Equifax. In the second, we will hear how Equifax was hacked.

 **Do This:** Click the audio symbol to play the podcast. Stop the podcast at 13:30.

Teaching Tip

Transcripts are provided in the Activity Guide for this lesson. You may want to have students follow along as they listen.

In this lesson, we only scratch the surface of the Equifax breach. You may want to listen further to the podcasts yourself, to be more informed of the issues. For example, after the breach was found and reported, Equifax tried to direct users to their own website to check if they had been affected. However, they ended up linking to a fake website!

If students don't understand how this applies to their own lives, you can make connections to their ability to buy a car in the future, or get a loan for an apartment.

 **Do This:** Click the audio symbol to play the podcast. Start the podcast at 6:05 and end at 11:13. **Note:** There is a swear word that occurs around a minute after we are stopping the podcast, so please make sure to stop early.

 **Discuss:** *What information does Equifax store? Why should I care?*

Discussion Goal: Equifax stores information on everyone who has a credit score. This information includes private data like social security numbers and whether or not you pay your bills on time. This information is often used when consumers want to make purchases and determines whether or not they are eligible and how high-interest rates

will be for loans. This can affect your family's ability to buy a house or a car.

 **Discuss:** *As a computing innovation what are the benefits of Equifax? What are the potential harms?*

Discussion Goal: *Benefits:* companies are able to use data to make decisions when deciding who to lend money to or how much a person can be trusted to pay back a loan. With this system, we can purchase large items on loan like cars or houses, which would be out of reach for many people if they had to pay outright. *Harms:* Decisions made are not always fair or equitable. In addition, sometimes information is incorrect and difficult to get changed. A lot of trust is put in a small number of companies who are making a profit-making these decisions.

 **Discuss:** *What are the security risks?*

Discussion Goal: Data that Equifax stores can be hacked and distributed for malicious purposes. With the personal information that Equifax stores, hackers can commit identity theft and make purchases, drain bank accounts, or ruin people's financial history.

Remarks

Equifax was a real-world system which was compromised by a software error and a human error. In this case, the software update would have prevented the hack.

One of the issue brought up with the Equifax Breach is that private data was stored, but there were no terms of service that users signed allowing that data to be collected. Equifax's customers are businesses who want to use that data to make decisions about people. Private citizens' data was and is constantly being collected - and potentially hacked. To summarize, Equifax sells information about private data that individual consumers did not sign terms of service to allow.

 **Discuss:** *What rules or regulations would you recommend be put in place to control how data is collected and shared? What role (if any) should the government play?*

Discussion Goal: This discussion is open-ended. There is no right or wrong answer here - prompt students to be thoughtful in the types of change they would like to see in how companies like Equifax are regulated.

Remarks

As a citizen, you have the power to bring about change. As more and more private information is knowingly or unknowingly collected and security risks continue to be a factor it's important to think about protecting what's ours and speaking to those who are in authority to strengthen protections. And someday, you yourself may be the person making these decisions and putting new laws in place!

Wrap Up (5 minutes)

 **Do This:** Run a quick [Kahoot quiz](#) to review Security Risks.

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: How does human error relate to security risks?





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